TCS Toolkit

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1 Asymptotics and Probability

1.1 Big O and friends

Recall that $f(x) = O(g(x))$ $(as\ x \to \infty) \iff \exists c, x_0\ s.t.\ |f(x)| \leq c \cdot g(x),\ \forall x \geq x_0$. **Example:** $\sum_{i=1}^{n} i = \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2} = O(n^2) = \frac{n^2}{2} + O(n)$.

Definition 1.1 (Harmonic Number). *Let $n \in \mathbf{N}$, then we define $H_n = 1 + 1/2 + 1/3 + 1/4 + \ldots + 1/n$.*

We can get an **upper bound** and a **lower bound** for harmonic number in the following way:

$$\begin{aligned} H_n = 1 + 1/2 + 1/3 + 1/4 + \ldots + 1/n \leq \\ 1 + 1/2 + 1/2 + 1/4 + 1/4 + 1/4 + 1/4 + 1/8 + \ldots \leq \\ \lfloor \log_2 n \rfloor + 1. \end{aligned}$$

$$\begin{aligned} H_n = 1 + 1/2 + 1/3 + 1/4 + \ldots + 1/n \geq \\ 1 + 1/2 + 1/4 + 1/4 + 1/8 + 1/8 + 1/8 + 1/8 + 1/16 \ldots \geq \\ 1/2\lceil \log_2 n \rceil + 1. \end{aligned}$$

Since that $H_n \leq O(\log n)$ and $H_n \geq \Omega(\log n)$, we can conclude that $H_n = \Theta(\log n)$.

Definition 1.2 ($f(x) \sim g(x)$). *We say that $f(x) \sim g(x)$ as $x \to \infty$ if $\frac{f(x)}{g(x)} \to 1$.*

We recall the definition of a Taylor Series:

Theorem 1.1 (Taylor Series). *If $f(x)$ is analytic at $x = a$, then it can be represented by its Taylor series:*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n,$$

which converges to $f(x)$ for all x within the radius of convergence.

Definition 1.3 (e^x). *We define the function $e^x = \sum_{i=0}^{\infty} \frac{x^n}{n!}$, $\forall x \in \mathbf{R}$.*

For example, the function $\ln 1 + x = \sum_{n=0}^{\infty}(-1)^{n-1}\frac{x^n}{n!}$, for $|x| \leq 1$. Additionally, we can use this equivalence to approximate $\ln 1 + x = \sum_{n=0}^{\infty}(-1)^{n-1}\frac{x^n}{n!} \sim x$ when $x \to 0^+$. We can also exponent this and get: $\exp x \sim 1 + x$.

Definition 1.4 (Central Binomial Coefficient). *We define the central binomial coefficient as $C_n := \binom{n}{n/2}$. We notice that the probability of getting half of n flipped coins is $\frac{C_n}{2^n}$.*

Proposition 1.2. $C_n = \Theta(\frac{2^n}{\sqrt{n}})$.

Proof. We can bound above and below the expression. Lets do a simple example to see this. $C_{10} = \frac{10 \cdot 9 \cdot 8 \cdots 2 \cdot 1}{5 \cdot 4 \cdots 1 \cdot 5 \cdot 4 \cdots 1}$. Then we divide by 2^n and get: $C_{10}/2^n = \frac{10 \cdot 9 \cdot 8 \cdots 2 \cdot 1}{10 \cdot 8 \cdots 2 \cdot 10 \cdot 8 \cdots 2}$. Furthermore we can potentiate to $()^2$ and get: $(\frac{C_n}{2^n})^2 = \frac{10 \cdot 10 \cdot 9 \cdot 9 \cdot 8 \cdot 8 \cdots 2 \cdot 2 \cdot 1}{10 \cdot 10 \cdot 10 \cdot 10 \cdot 8 \cdot 8 \cdot 8 \cdot 8 \cdots 2 \cdot 2 \cdot 2 \cdot 2} = \frac{9 \cdot 9 \cdot 7 \cdot 7 \cdot 5 \cdot 5 \cdot 3 \cdot 3}{10 \cdot 10 \cdot 8 \cdot 8 \cdot 6 \cdot 6 \cdot 4 \cdot 4 \cdot 2 \cdot 2}$. From this we can get an upper bound: $(\frac{C_n}{2^n})^2 \leq \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2} = \frac{1}{11} = \frac{1}{n+1}$. Also we can get a lower bound: $(\frac{C_n}{2^n})^2 \geq \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{1}{10} = \frac{1}{n}$. $\square$

1.2 Sum of Random Variables, Gaussian and Central Limit Theorem

We work with the following setup: let $X_1, X_2, ..., X_n$ be i.i.d random variables with $\mathbf{P}[X_i = 1] = p$ and $\mathbf{P}[X_i = 0] = 1 - p = q$. Let $S_n = X_1 + X_2 + ... + X_n$, then $\mathbf{E}(S_n) = \mathbf{E}(X_1) + ... + \mathbf{E}(X_n) = pn$.

2 Computational Models

2.1 Circuits

Definition 2.1 (Circuit)**.** *A circuit is a Directed Acyclic Graphs (DAGs) that computes Boolean Functions $f : \{0,1\}^n \to \{0,1\}$. The nodes of a circuit are called gates. A circuit can be seen as a tree, where the leafs are input gates that recieves an input $x = (x_1, x_2, ..., x_n)$ and the root generates an output bit.*

Definition 2.2 (Size and Depth of Circuit)**.** *The Size and Depth of a circuit C are defined as the number of non-input gates and the longest input-output path, respectively.*

Example 2.1. *Some common basis are the following: $\mathcal{B}_1 = \{\neg, binary \vee, binary \wedge\}, \mathcal{B}_2 = \{all\ g : \{0,1\}^2 \to \{0,1\}\}, \mathcal{B}_3 = \{\neg, all\ fan\text{-}in \vee, all\ fan\text{-}in \wedge\}$. Notice that $\mathcal{B}_3$ allow for constant depth computations. For example, we can add two n-bits numbers in constant depth. Instead for B_1, we need at least a depth of $\log n$ to look all bits.*

Example 2.2. *We study the function **AND**: $\{0,1\}^n \to \{0,1\}$. Notice that with $\mathcal{B}_3$ we can generate a circuit with size 1. Instead with $\mathcal{B}_2$ we can generate a circuit with size $O(n)$ and depth $O(\log n)$.*

Example 2.3. *We study the **Palindrome** function $p : \{0,1\}^{2n} \to \{0,1\}$. With $\mathcal{B}_3$ we can generate a circuit with depth 2, by generating an "equal" gate.*

Definition 2.3 (Basis)**.** *We call $\mathcal{B}(C)$ the basis of a circuit C, which is a set of allowed operations in gates of a circuit C.*

We have talk about constant size inputs of size n, this mean that for each input size we need to generate a different circuit. In order to accept a certain language L, we define a **circuit family** $(C_n)_{n=1}^{\infty}$, where the circuit C_n should accepts all words in $L \cap \{0,1\}^n$. Now, we define some families of languages that are categorized by the size of circuit families (with $\mathcal{B}_3$) that accept them.

Definition 2.4 (Family AC^0)**.** *The collection of languages decided by a circuit family of depth $O(1)$ and size $poly(n)$.*

Definition 2.5 (Family NC)**.** *The collection of languages decided by a circuit family of depth $polylog(n)$ and size $poly(n)$.*